TOPIC 3: FINANCIAL MANAGEMENT • HIGH-PRIORITY EXAM PRACTICE

# Extra Exam Problems: Circonsa, Infopymes, Wifinet, and Cemed

**Official Source Document:** `T3_Additional_EXERCISES.pdf` (UC3M Aerospace Engineering) • Step-by-step exam strategy, analytical deductions, and balanced financial statements.

## Method: building the income statement and the balance sheet

**Central idea:** every operation of an exam problem is read **three times**: does it create a **revenue or expense** of the period (income statement)? does **cash** enter or leave before the closing date (cash account)? what **asset, liability or equity** item remains open at the closing date (balance sheet)? If the three readings are done for every operation, the balance sheet closes by itself.

### Step 0. Read the dates: the period and the closing instant

- Write the **period** of the income statement (a quarter, a semester, a calendar year) and the **closing instant** of the balance sheet (e.g. 31/12 at 24:00). The income statement covers a **period**; the balance sheet is a **point in time** [Slide 7, Slide 12].
- Draw a **timeline** with every dated event: purchases, sales, payments, loan dates, repayments. Months of use inside the period decide depreciation and interest; payment dates decide cash and debts.
- A balance at "24:00 after all payments" includes the payments made that day (taxes, dividends, instalments); a balance "before" them keeps them as liabilities.

### Step 1. Classify every operation with the three questions

For each operation fill one row of this table:

| Operation | Income statement (revenue / expense of the period) | Cash (collected / paid before closing) | Balance sheet item still open at closing |
|---|---|---|---|
| Sale on credit | Revenue for the **whole** sale [Slide 8] | Only the part collected | Accounts receivable = sale − collected |
| Purchase of goods for resale or raw materials | Expense only for the part **consumed or sold** (cost of goods sold) [Slide 10] | Only the part paid | Inventory = unconsumed part; accounts payable = unpaid part |
| Salaries, utilities, rent, services | Expense for the period's consumption [Slide 8] | Only the part paid | Payable for the unpaid part |
| Purchase of a fixed asset (machine, building, patent) | No expense on purchase; only the **depreciation** of the months used | Only the part paid | Asset at cost minus accumulated depreciation; debt for the unpaid part |

| Operation | Income statement (revenue / expense of the period) | Cash (collected / paid before closing) | Balance sheet item still open at closing |
|---|---|---|---|
| Loan received | Nothing (it is not a revenue) | Cash in | Liability: current part (due within one year) and long-term part [Slide 16] |
| Interest on a loan | Expense for the months of the period (non-operating item) [Slide 11] | Only the interest actually paid | Interest payable for the accrued but unpaid part |
| Repayment of principal | Nothing (it is not an expense) | Cash out | The liability decreases |
| Capital contributed by owners | Nothing | Cash in (or the asset contributed) | Contributed capital [Slide 17] |
| Income taxes | Expense = income before taxes × rate [Slide 11] | Only if paid before closing | Taxes payable otherwise |
| Dividends | Nothing (a distribution of profit, not an expense) [Slide 23] | Only if paid before closing | Dividends payable if declared and unpaid; retained earnings = profit − dividends |

Two consequences are worth memorising: **a revenue is not a collection and an expense is not a payment** [Slide 8]; and **depreciation is an expense that is never a payment**, while **buying a machine or repaying a loan are payments that are never expenses**.

### Step 2. Compute the special items

**Depreciation** (when the statement says linear / straight-line, with or without residual value):

$$\text{Depreciation of the period} = \frac{\text{Cost} - \text{Residual value}}{\text{Useful life in years}} \times \frac{\text{months used in the period}}{12}$$

The asset appears in the balance sheet at its **net value**: cost − accumulated depreciation. Land is not depreciated in the exercises (no useful life is given for it).

**Interest** (simple interest at an annual rate, as in all the exercises):

$$\text{Interest of the period} = \text{Principal outstanding} \times \text{annual rate} \times \frac{\text{months}}{12}$$

If the principal changes during the period (instalments), split the period into the stretches where the principal is constant and add the pieces.

**Cost of goods sold** with stocks:

$$\text{COGS} = \text{Initial inventory} + \text{Purchases} - \text{Ending inventory}$$

This is the same rule as "expense only for the part consumed": whatever was bought or held and is no longer in stock was consumed.

### Step 3. Build the income statement (format of [Slide 9])

| Line | Content |
|---|---|
| Sales revenue | Sales of the period (net of the discounts the statement gives) |
| − Cost of goods sold | COGS of Step 2 |
| = Gross profit | |
| − Operating expenses | Salaries, social security, rent, utilities, services, depreciation |
| = Income from operations | |
| ± Non-operating items | Interest expense and interest revenue; gains and losses on the sale of equipment and investments [Slide 11] |
| = Income before taxes | |
| − Income taxes | Income before taxes × tax rate [Slide 11] |
| = Net income | Then split it: dividends and retained earnings [Slide 23] |

### Step 4. Compute cash with a cash account

$$\text{Cash}_{\text{end}} = \text{Cash}_{\text{start}} + \sum \text{collections} - \sum \text{payments}$$

List every inflow (capital, loans, collections from customers, old receivables collected) and every outflow (purchases paid, salaries, investments paid, interest, principal, taxes, dividends). Depreciation never appears here.

### Step 5. Build the balance sheet (format of [Slide 18])

| Assets | Liabilities and equity |
|---|---|
| **Non-current assets:** land, buildings, machinery, equipment, patents, licences (at net value); long-term investments; receivables due in more than one year [Slide 15] | **Equity:** contributed capital; retained earnings / reserves; net income of the period not distributed [Slide 17] |
| **Current assets:** inventory, accounts receivable due within one year, cash [Slide 14] | **Long-term liabilities:** debts due after one year [Slide 16] |

| Assets | Liabilities and equity |
|---|---|
| | **Current liabilities:** suppliers, other payables, taxes payable, interest payable, dividends payable, the part of any loan due within one year [Slide 16] |

The one-year boundary is counted **from the balance-sheet date**: a debt due on 1/2 of next year is current at 31/12; a customer debt deferred to July of the year after next is non-current.

### Step 6. Check

1. Total assets $=$ Total liabilities $+$ Equity [Slide 12]. If it fails, the difference is almost always one operation that was recorded on only one side.
2. **Cash cross-check:** the change in equity due to the period equals net income − dividends; the change in cash equals collections − payments. Re-adding each operation should explain both.
3. **Sign and size:** receivables, payables and inventory can never be negative; depreciation can never exceed cost − residual value.

`Additional Exercise 1 · Exam type`

## T3 Additional Exercise 1 — CIRCONSA, statements at 30 June 2015

`Original statement · T3_Additional_EXERCISES.pdf`

The financial director of CIRCONSA is working on its financial statements (balance sheet and income statement) as of 30 June 2015. It has the balance of the company as of 31 December 2014 and the following operations that took place from 1 January 2015 to 30 June 2015. The company sells a certain product.

- **i.** During the semester the company sold 1,200 units at 20 m.u./unit (every month the same amount, 200 u.). Customers pay 30 days after the sale. A very important customer (30% of the sales), who bought an equal amount each month, was given a 10% discount on its purchases.
- **ii.** During the semester 1,000 units were purchased (40% in the first quarter and 60% in the second) at 12 m.u./u. Suppliers are paid at 90 days.
- **iii.** At 31/12/2014 there is a stock of 200 units valued at 12 m.u./u.
- **iv.** During the first quarter of 2015 the taxes and the outstanding supplier balances of the previous year were paid.
- **v.** During the first quarter of 2015 the customer debt outstanding from the previous year was collected, except 300 m.u. from a customer who requested to defer payment until July 2016.

- **vi.** Personnel expenses were 350 m.u./month (plus 150 for Social Security) and general expenses 300 m.u./month. Personnel and Social Security are paid in the month in which they occur; overheads two months after.
- **vii.** On 1/04/2015 a renting contract (3 years) for computer equipment was signed with a specialised company, 400 m.u. per month, with payment at 120 days (payment at the end of each month).
- **viii.** On 1/04/2015 a patent was bought for 2,400 m.u., to be paid on 1/07/2015. It is amortised in 4 years.
- **ix.** On 1/02/2015 a credit of 2,000 m.u. was obtained, to be returned a year later with 12% annual interest; interest and principal are paid on 1/02/2016.
- **x.** At 31/12/2014 the company had only one machine, bought on 1/07/2014, amortised linearly with a residual value of 20%.
- **xi.** On 1/02/2015 a capital increase (fully paid, no issue premium) of 10,000 m.u. was made. The same day a new machine costing 12,000 m.u. was bought: 8,000 paid in cash and the remainder on 1/07/2015. It is amortised in 4 years with zero residual value.
- **xii.** Assume a 20% corporate tax, recorded in the balance at 30/06/2015 as "taxes to pay".

**Balance sheet at 31/12/2014 (m.u.):** current assets 5,400 (cash 1,200; accounts receivable 1,800; inventory 2,400); non-current assets 450 (machinery 500; accumulated depreciation 50); current liabilities 800 (unpaid taxes 350; suppliers 450); owners' equity 5,050 (retained earnings 850; paid-in capital 4,200); total 5,850.

**Question:** calculate the balance sheet and the profit and loss account of CIRCONSA at 30 June 2015.

### Phase 1 – Setup: data, timeline and hypotheses

**Period:** 1/01/2015 to 30/06/2015, **6 months**. **Closing instant:** 30/06/2015. Every monthly figure is multiplied by the months it applies inside this window.

| Date | Event |
|---|---|
| Jan-Mar | Sales 200 u./month; purchases 400 u. (Q1); payment of 2014 taxes and suppliers; collection of 2014 receivables except 300 |
| 1/02 | Credit 2,000 received; capital increase 10,000; machine 2 bought (8,000 paid) |
| 1/04 | Renting starts (400/month); patent bought (2,400, payable 1/07) |
| Apr-Jun | Sales 200 u./month; purchases 600 u. (Q2) |
| 30/06 | **Closing** |

| Date | Event |
|---|---|
| 1/07 | Payment of patent and of the 4,000 left on machine 2 (after the closing) |

**Hypotheses** (all read from the statement):

1. Sales are uniform (200 u./month) and collected 30 days later, so at 30/06 only **June's sales** are uncollected.
2. Q1 purchases are paid 90 days later, inside the semester; **Q2 purchases** are still unpaid at 30/06.
3. Overheads are paid two months later: January-April are paid, **May and June** are unpaid.
4. The renting is paid at 120 days: none of the April-June instalments has been paid at 30/06.
5. Simple interest, prorated by months. Tax: 20% of income before taxes, left unpaid.
6. Machine 1: its useful life is not given; it is deduced from the opening balance (Phase 3, Step 5).

## Phase 2 – Frames: classification of every operation

The three questions of the method (income statement? cash? open balance item at 30/06?) applied to each operation:

| Op. | Income statement (6 months) | Cash in (+) / out (–) | Open at 30/06 |
|---|---|---|---|
| i. Sales | Net sales 23,280 | +19,400 (Jan-May) | Receivables 3,880 (June) |
| ii-iii. Purchases and stock | COGS 14,400 | –4,800 (Q1) | Inventory 0; suppliers 7,200 (Q2) |
| iv. 2014 taxes and suppliers | – | –350 – 450 | – |
| v. 2014 receivables | – | +1,500 | Long-term receivable 300 |
| vi. Personnel + SS | 3,000 | –3,000 | – |
| vi. Overheads | 1,800 | –1,200 (Jan-Apr) | Overheads payable 600 |
| vii. Renting | 1,200 | 0 | Renting payable 1,200 |
| viii. Patent | Amortisation 150 | 0 | Patent 2,400 – 150; patent payable 2,400 |
| ix. Credit | Interest 100 | +2,000 | Credit 2,000; interest payable 100 |
| x. Machine 1 | Depreciation 50 | 0 | Machine 500 – 100 |
| xi. Capital increase | – | +10,000 | Capital +10,000 |
| xi. Machine 2 | Depreciation 1,250 | –8,000 | Machine 12,000 – 1,250; debt 4,000 |
| xii. Tax | 266 | 0 | Taxes payable 266 |

Each figure is derived in Phase 3.

## Phase 3 – Derivation

**Step 1. Sales and receivables (op. i)**

Gross sales per month: $200 \times 20 = 4{,}000$ m.u. The important customer buys 30% of them, so its discount per month is

$$0.30 \times 4{,}000 \times 0.10 = 120 \text{ m.u./month}$$

Net sales per month: $4{,}000 - 120 = 3{,}880$ m.u. Over the semester:

$$\text{Net sales} = 6 \times 3{,}880 = 23{,}280 \text{ m.u.} \quad (= 24{,}000 - 720)$$

Collections at 30 days: the sales of January to May are collected by 30/06 ($5 \times 3{,}880 = 19{,}400$); June's sales are still owed: **receivables 3,880**.

**Step 2. Purchases, stock and cost of goods sold (ops. ii, iii)**

Units: initial stock 200 + purchases 1,000 = 1,200 available; 1,200 sold, so the ending stock is 0 units. All units cost 12 m.u., so

$$\text{COGS} = \text{Initial inventory} + \text{Purchases} - \text{Ending inventory} = 2{,}400 + 12{,}000 - 0 = 14{,}400 \text{ m.u.}$$

Purchases: Q1 $= 0.40 \times 1{,}000 \times 12 = 4{,}800$, paid at 90 days (inside the semester); Q2 $= 0.60 \times 1{,}000 \times 12 = 7{,}200$, **unpaid at 30/06**.

**Step 3. Items from 2014 (ops. iv, v)**

Paid in Q1: taxes 350 and suppliers 450 (cash –800; both liabilities disappear). Collected in Q1: $1{,}800 - 300 = 1{,}500$. The remaining 300 is due in July 2016, more than one year after 30/06/2015, so it becomes a **non-current receivable** [Slide 15].

None of these affects the income statement: they settle revenues and expenses of 2014.

**Step 4. Personnel, overheads and renting (ops. vi, vii)**

- Personnel + Social Security: $6 \times (350 + 150) = 3{,}000$, all paid.
- Overheads: $6 \times 300 = 1{,}800$ of expense; paid two months later, so January-April are paid ($4 \times 300 = 1{,}200$) and May-June are owed: **600 payable**.
- Renting: April, May, June, $3 \times 400 = 1{,}200$ of expense; paid at 120 days, nothing paid by 30/06: **1,200 payable**.

**Step 5. Fixed assets: depreciation and amortisation (ops. viii, x, xi)**

**Machine 1.** Cost 500, residual value $0.20 \times 500 = 100$, depreciable base $500 - 100 = 400$. The opening accumulated depreciation is 50 for the 6 months from 1/07/2014 to 31/12/2014, so the annual depreciation is $50 \times \frac{12}{6} = 100$ and the useful life is $\frac{400}{100} = 4$ years. For the semester:

$$D_1 = \frac{500 - 100}{4} \times \frac{6}{12} = 50 \text{ m.u.}, \qquad \text{accumulated} = 50 + 50 = 100$$

**Machine 2.** Bought 1/02, used 5 months (February-June), zero residual value:

$$D_2 = \frac{12{,}000 - 0}{4} \times \frac{5}{12} = 1{,}250 \text{ m.u.}$$

**Patent.** Bought 1/04, used 3 months:

$$A = \frac{2{,}400}{4} \times \frac{3}{12} = 150 \text{ m.u.}$$

Total depreciation and amortisation: $50 + 1{,}250 + 150 = 1{,}450$ m.u. None of it is a payment.

Payments: machine 2, 8,000 paid in cash; the other 4,000 and the whole patent (2,400) are due on 1/07/2015, one day after the closing, so they are **current liabilities** at 30/06.

**Step 6. Credit and interest (op. ix)**

Received 1/02 (cash +2,000), returned on 1/02/2016, within one year of 30/06/2015: **current liability**. Interest from 1/02 to 30/06 (5 months), paid only in 2016:

$$I = 2{,}000 \times 0.12 \times \frac{5}{12} = 100 \text{ m.u.} \quad \text{(expense and interest payable)}$$

**Step 7. Income statement for the semester**

| Line | Computation | m.u. |
|---|---|---|
| Net sales | 24,000 − 720 | 23,280 |
| − Cost of goods sold | Step 2 | 14,400 |
| **= Gross profit** | 23,280 − 14,400 | **8,880** |
| − Personnel and Social Security | | 3,000 |

| Line | Computation | m.u. |
|---|---|---|
| – Overheads | | 1,800 |
| – Renting | | 1,200 |
| – Depreciation and amortisation | $50 + 1{,}250 + 150$ | 1,450 |
| **= Income from operations** | $8{,}880 - 7{,}450$ | **1,430** |
| – Interest expense | Step 6 | 100 |
| **= Income before taxes** | $1{,}430 - 100$ | **1,330** |
| – Income taxes (20%) | $0.20 \times 1{,}330$ | 266 |
| **= Net income** | $1{,}330 - 266$ | **1,064** |

### Step 8. Cash at 30/06/2015

| Inflows | m.u. | Outflows | m.u. |
|---|---|---|---|
| Opening cash | 1,200 | 2014 taxes | 350 |
| 2014 receivables collected | 1,500 | 2014 suppliers | 450 |
| Sales Jan-May collected | 19,400 | Q1 purchases | 4,800 |
| Credit | 2,000 | Personnel + SS | 3,000 |
| Capital increase | 10,000 | Overheads Jan-Apr | 1,200 |
| | | Machine 2 (cash part) | 8,000 |
| **Total** | **34,100** | **Total** | **17,800** |

$$\text{Cash}_{30/06} = 34{,}100 - 17{,}800 = 16{,}300 \text{ m.u.}$$

### Step 9. Balance sheet at 30/06/2015

| Assets | m.u. | Liabilities and equity | m.u. |
|---|---|---|---|
| Machinery (cost $500 + 12{,}000$) | 12,500 | Paid-in capital ($4{,}200 + 10{,}000$) | 14,200 |
| – Accumulated depreciation ($100 + 1{,}250$) | (1,350) | Retained earnings (2014) | 850 |
| Patent ($2{,}400 - 150$) | 2,250 | Net income of the semester | 1,064 |
| Long-term receivable (due July 2016) | 300 | **Equity** | **16,114** |
| **Non-current assets** | **13,700** | Credit (due 1/02/2016) | 2,000 |
| Inventory | 0 | Interest payable | 100 |
| Accounts receivable (June) | 3,880 | Suppliers (Q2 purchases) | 7,200 |
| Cash | 16,300 | Debt for machine 2 (due 1/07) | 4,000 |
| **Current assets** | **20,180** | Debt for patent (due 1/07) | 2,400 |

| Assets | m.u. | Liabilities and equity | m.u. |
|---|---|---|---|
| | | Overheads payable | 600 |
| | | Renting payable | 1,200 |
| | | Taxes payable | 266 |
| | | **Current liabilities** | **17,766** |
| **Total assets** | **33,880** | **Total liabilities and equity** | **33,880** |

There are no long-term liabilities: every debt is due before 30/06/2016.

## Phase 4 – Interpretation and checks

- **Balance:** $13{,}700 + 20{,}180 = 33{,}880 = 16{,}114 + 17{,}766$.
- **Equity bridge:** equity grows from 5,050 to 16,114, that is $+11{,}064 = 10{,}000$ (capital increase) $+1{,}064$ (net income).
- **Profit is not cash:** the semester earns 1,064 but cash grows by $16{,}300 - 1{,}200 = 15{,}100$, mostly because of the capital increase (10,000), the credit (2,000) and the many expenses still unpaid (suppliers, renting, overheads, interest, taxes). In the opposite direction, 8,000 of cash went into a machine whose expense this semester is only 1,250.
- **Limiting case:** if June's sales had been collected, cash would be 3,880 higher and receivables zero, but net income would not change: revenue is recognised when the sale is made [Slide 8].
- **Exam traps:** (1) the 1/07 debts are liabilities at 30/06, not payments; (2) the 300 deferred to July 2016 is non-current; (3) machine 2 is depreciated 5 months and the patent 3 months, not 6; (4) the life of machine 1 must be deduced from the opening balance.

## Check against the official key

OFFICIAL SOLUTION, T3_ADDITIONAL_SOLUTIONS.PDF, PP. 6-8

The income statement and the cash account coincide with the key: net sales 23,280; cost of goods sold 14,400; personnel 3,000; general expenses 1,800; renting 1,200; EBITDA 2,880; depreciation and amortisation 1,450; EBIT 1,430; interest 100; income before taxes 1,330; tax 266; net income 1,064; collections 34,100; cash 16,300; total 33,880. The key writes the depreciation of machine 1 as $(500 - 100)/8$, that is, 8 half-years, the same 4-year life deduced above.

DISCREPANCY WITH THE OFFICIAL SOLUTION (THE 300 DEFERRED TO JULY 2016)

The key shows accounts receivable of 4,180, which includes the 300 due in July 2016, so it counts them as a current asset (current assets 20,480; non-current assets 13,400). Here they are a non-current receivable, because they fall due more than one year after 30/06/2015 (Slide 15), so receivables are 3,880 + 300 (current assets 20,180; non-current assets 13,700). Total assets, liabilities and equity are the same in both.

**ERRATUM IN THE OFFICIAL SOLUTION**

The key's payments column adds to 16,800, but its items add to $450 + 3{,}000 + 1{,}200 + 4{,}800 + 350 + 8{,}000 = 17{,}800$, which gives the cash of 16,300 that it does show. The capital increase in the equity block reads 1,000 instead of 10,000 (only 10,000 gives equity of 16,114), and the labels of the interest payable (100) and the tax payable (266) are swapped.

Additional Exercise 2 · Exam type

# T3 Additional Exercise 2 – INFOPYMES, statements 2016-2017 and ROA/ROE

Original statement · T3_Additional_EXERCISES.pdf, p. 3

Three entrepreneurs established INFOPYMES on 1/01/2016 to provide on-line information services for small and medium enterprises, start-ups and potential entrepreneurs.

- The computer equipment, worth 100,000 €, was acquired on 1/01/2016 and will be paid to suppliers after three months, on 1/04/2016. Useful life 5 years (until 31/12/2020), linear depreciation, no residual value.
- On-line consultations brought revenues of 300,000 € in 2016.
- Rented premises: 1,000 €/month from 1/01/2016. Salaries and Social Security of the permanent employees: 5,000 €/month. Electricity, phone, etc.: 500 €/month.
- Variable wages of the additional staff for the on-line consultations: 174,000 € in 2016.
- The partners contributed an initial capital of 100,000 €. A bank granted a loan of 100,000 € at 10% yearly interest for 5 years (1/01/2016 to 31/12/2020). Each 31 December the company returns 20,000 € and pays the interest (10%) on the loan value at the beginning of each year.
- Results are declared at 23:59 on 31 December (after all payments and receipts, including interest, wages, Social Security, etc.), and the corporation tax of 30% on the profits is paid immediately. Dividends of 40% of the profits after taxes are distributed, and the final balance is drawn at 24:00.

**Required:** (1) the income statement for 2016; (2) assuming the tax has been paid and the dividends distributed, the balance sheet at 24:00 on 31/12/2016; (3) the balance sheet at 24:00 on 31/12/2017, assuming no new investments in 2017 and that the retained earnings and the cash balance at the end of 2017 remained unchanged with respect to 2016; (4) the ROA and ROE ratios in 2017.

## Phase 1 – Setup: data, timeline and hypotheses

| Item | Value |
|---|---|
| Capital | 100,000 € (1/01/2016) |
| Loan | 100,000 € at 10%; 20,000 € returned each 31/12; interest on the balance at the start of each year |
| Equipment | 100,000 €, bought 1/01/2016, paid 1/04/2016, 5 years, no residual value |
| Revenues 2016 | 300,000 € |

| Item | Value |
|---|---|
| Fixed monthly costs | Rent 1,000; permanent staff 5,000; utilities 500 (€/month) |
| Variable wages 2016 | 174,000 € |
| Tax | 30% of profits, paid at 23:59 on 31/12 |
| Dividends | 40% of profit after taxes, paid at 23:59 on 31/12 |

**Hypotheses:**

1. The statement says the results are declared "after having made all payments and receipts", so the 300,000 € of revenues are **collected** and all 2016 costs are **paid** within 2016.
2. The balance is drawn at 24:00, **after** paying tax, interest, the 20,000 € instalment and dividends: none of them remains as a liability.
3. For 2017 the statement gives no operating data; we assume the **same revenues and operating costs as in 2016**, which is the only reading that lets requirement 4 be computed. With retained earnings unchanged, the whole 2017 profit must have been paid out as dividends (shown in Phase 3).

## Phase 2 – Frames: classification of every operation (2016)

| Operation | Income statement 2016 | Cash 2016 | Open at 24:00 on 31/12/2016 |
|---|---|---|---|
| Capital | – | +100,000 | Capital 100,000 |
| Loan | – | +100,000 | Loan 80,000 (after the instalment) |
| Equipment | Depreciation 20,000 | −100,000 (on 1/04) | Equipment 100,000 − 20,000 |
| Revenues | 300,000 | +300,000 | – |
| Rent, staff, utilities, variable wages | 252,000 | −252,000 | – |
| Interest | 10,000 | −10,000 | – |
| Instalment | – | −20,000 | (reduces the loan) |
| Tax | 5,400 | −5,400 | – |
| Dividends | – (distribution) | −5,040 | Retained earnings 7,560 |

The equipment is bought on credit on 1/01 and paid on 1/04: between those dates it is a debt to the supplier, but at 31/12 that debt no longer exists.

## Phase 3 – Derivation

### Step 1. Depreciation and interest 2016

$$D = \frac{100{,}000 - 0}{5} \times \frac{12}{12} = 20{,}000\ €/\text{year}$$

$$I_{2016} = 100{,}000 \times 0.10 = 10{,}000\ € \quad \text{(loan value at the beginning of 2016)}$$

### Step 2. Income statement 2016 (requirement 1)

| **Line** | **Computation** | **€** |
|---|---|---|
| Revenues (on-line consultations) | | 300,000 |
| – Rent | $12 \times 1{,}000$ | 12,000 |
| – Permanent staff and Social Security | $12 \times 5{,}000$ | 60,000 |
| – Electricity, phone, etc. | $12 \times 500$ | 6,000 |
| – Variable wages | | 174,000 |
| – Depreciation | Step 1 | 20,000 |
| **= Income from operations** | $300{,}000 - 272{,}000$ | **28,000** |
| – Interest expense | Step 1 | 10,000 |
| **= Income before taxes** | $28{,}000 - 10{,}000$ | **18,000** |
| – Income taxes (30%) | $0.30 \times 18{,}000$ | 5,400 |
| **= Net income** | $18{,}000 - 5{,}400$ | **12,600** |

Distribution [Slide 23]: dividends $= 0.40 \times 12{,}600 = 5{,}040\ €$; retained earnings (self-financing) $= 12{,}600 - 5{,}040 = 7{,}560\ €$.

### Step 3. Cash at 24:00 on 31/12/2016

| **Inflows** | **€** | **Outflows** | **€** |
|---|---|---|---|
| Capital | 100,000 | Equipment (1/04) | 100,000 |
| Loan | 100,000 | Rent + staff + utilities | 78,000 |
| Revenues collected | 300,000 | Variable wages | 174,000 |
| | | Interest | 10,000 |
| | | Loan instalment | 20,000 |

| Inflows | € | Outflows | € |
|---|---|---|---|
| | | Tax | 5,400 |
| | | Dividends | 5,040 |
| **Total** | **500,000** | **Total** | **392,440** |

$$\text{Cash}_{2016} = 500{,}000 - 392{,}440 = 107{,}560\,€$$

**Step 4. Balance sheet at 24:00 on 31/12/2016 (requirement 2)**

The loan after the first instalment is $100{,}000 - 20{,}000 = 80{,}000$ €. The instalment due on 31/12/2017 (20,000 €) is within one year, so it is a **current liability**; the other 60,000 € are long-term [Slide 16].

| Assets | € | Liabilities and equity | € |
|---|---|---|---|
| Equipment | 100,000 | Capital | 100,000 |
| – Accumulated depreciation | (20,000) | Retained earnings | 7,560 |
| **Non-current assets** | **80,000** | **Equity** | **107,560** |
| Cash | 107,560 | Long-term bank loan | 60,000 |
| **Current assets** | **107,560** | Short-term bank loan (due 31/12/2017) | 20,000 |
| **Total assets** | **187,560** | **Total liabilities and equity** | **187,560** |

**Step 5. Balance sheet at 24:00 on 31/12/2017 (requirement 3)**

Item by item:

- **Equipment:** no new investment; two years of depreciation, $2 \times 20{,}000 = 40{,}000$; net value $100{,}000 - 40{,}000 = 60{,}000$ €.
- **Loan:** second instalment paid, $80{,}000 - 20{,}000 = 60{,}000$ €: 20,000 € current (due 31/12/2018) and 40,000 € long-term.
- **Cash and retained earnings:** unchanged by statement: 107,560 € and 7,560 €.
- **Capital:** unchanged, 100,000 €.

| Assets | € | Liabilities and equity | € |
|---|---|---|---|
| Equipment | 100,000 | Capital | 100,000 |
| – Accumulated depreciation | (40,000) | Retained earnings | 7,560 |
| **Non-current assets** | **60,000** | **Equity** | **107,560** |
| Cash | 107,560 | Long-term bank loan | 40,000 |
| **Current assets** | **107,560** | Short-term bank loan (due 31/12/2018) | 20,000 |
| **Total assets** | **167,560** | **Total liabilities and equity** | **167,560** |

**Is the statement consistent?** With the 2016 operations repeated (hypothesis 3), 2017 gives:

$$I_{2017} = 80{,}000 \times 0.10 = 8{,}000, \quad \text{IBT} = 28{,}000 - 8{,}000 = 20{,}000, \quad T = 0.30 \times 20{,}000 = 6{,}000, \quad \text{NI} = 14{,}000$$

Retained earnings unchanged means dividends $= 14{,}000$ € (a 100% pay-out). Cash then moves by $+300{,}000 - 252{,}000 - 8{,}000 - 20{,}000 - 6{,}000 - 14{,}000 = 0$, so the cash is indeed unchanged: the two conditions of the statement are compatible.

**Step 6. ROA and ROE in 2017 (requirement 4)**

> **DEFINITIONS NOT IN THE TOPIC 3 SLIDES**
> The Topic 3 deck in `sources/` does not define ROA and ROE, but the problem asks for them. The standard definitions used here are $\text{ROA} = \dfrac{\text{Income from operations}}{\text{Total assets}}$ (the return of the economic structure) and $\text{ROE} = \dfrac{\text{Net income}}{\text{Equity}}$ (the return to the owners). Check them against your class notes: if your lecturer defines ROA with net income instead, the value is $14{,}000/167{,}560 = 8.36\%$.

$$\text{ROA}_{2017} = \frac{28{,}000}{167{,}560} = 0.1671 = 16.71\%$$

$$\text{ROE}_{2017} = \frac{14{,}000}{107{,}560} = 0.1302 = 13.02\%$$

## Phase 4 – Interpretation and checks

- **Balances:** $80{,}000 + 107{,}560 = 187{,}560 = 107{,}560 + 60{,}000 + 20{,}000$ (2016) and $60{,}000 + 107{,}560 = 167{,}560 = 107{,}560 + 40{,}000 + 20{,}000$ (2017).
- **What changes between the two years:** both sides shrink by 20,000 €: the asset loses 20,000 € of value through depreciation, and the loan falls by the 20,000 € instalment, which in cash terms was financed by the cash generated (depreciation is an expense that is not a payment).
- **Leverage** [Slide 36]: $L/E = 80{,}000/107{,}560 = 0.74$ in 2016 and $60{,}000/107{,}560 = 0.56$ in 2017; the company becomes less indebted each year.
- **ROE < ROA here:** the return to the owners is lower than the return of the assets because interest and taxes are deducted from the numerator; the equity base being smaller than total assets is not enough to compensate.

- **Exam traps:** (1) the loan instalment is a payment, never an expense; (2) at 24:00 no tax or dividend liability remains; (3) the current part of the loan (next instalment) must be separated from the long-term part.

### Check against the official key

OFFICIAL SOLUTION, T3_ADDITIONAL_SOLUTIONS.PDF, PP. 12-14

Requirements 1 and 2 coincide with the key in every figure: EBIT 28,000; interest 10,000; income before taxes 18,000; tax 5,400; net income 12,600; dividends 5,040; retained earnings 7,560; cash 107,560; equipment net 80,000; total 187,560; bank loan 20,000 current and 60,000 long term; equity 107,560.

The key does not solve requirement 3 (balance sheet at 31/12/2017) nor requirement 4 (ROA and ROE), so Steps 5 and 6 above are not covered by it.

Additional Exercise 3 · Exam type

# T3 Additional Exercise 3 – WIFINET S.L., statements 2017 and dividend options

Original statement · T3_Additional_EXERCISES.pdf, p. 4

Wifinet S.L. was incorporated on 1/01/2016 with a registered capital of 6,003 m.u., to provide wireless Internet on the Alicante coast.

**Fiscal year 2016.** It took a bank loan of 2,000 m.u. at 5% annual interest (payable at the end of each year), the principal being repayable in 2019. It invested 5,371 m.u. in technical equipment with a life of 20 years; at the end of 2016 it recorded an accumulated depreciation of 26 m.u. for those assets. It also invested 206 m.u. in licence fees by 31/12/2016, to use certain technologies for 5 years. It spent 636 m.u. on financial shares of other companies. Most of 2016 was devoted to building the network and business started only in the last two months. At the end of 2016 the balance sheet reflected a **loss of 340 m.u.** and **cash in hand and bank of 1,476 m.u.**

In 2017 the results were positive, but the company decided not to distribute dividends and to retain all its profits to reduce the loss of the previous year.

**Fiscal year 2017.**

- Internet and VOIP: 5,000 customers on average, with an average revenue of 0.032 m.u./month per client in subscription fees. Additionally, 46 m.u. of on-line sales of other products. At the end of 2017 customers owed 164 m.u.
- Stocks were exhausted at the end of 2016. Purchases of materials to be installed in customers' homes: 680 m.u., of which 560 m.u. paid immediately and the rest deferred to the following year. The 2017 sales consumed half of the purchased goods.
- Staff costs: 559 m.u., paid in full in 2017. Supplies and outside services: 312 m.u., paid.
- Other administrative expenses: 300 m.u., 60% of them deferred to the following year.
- Corporation tax: 30%, payable in the first quarter of 2018.

**Required:** (1) the income statement for 2017; (2) the balance sheet at 31/12/2017; (3) which balance-sheet items change, and by how much, if a dividend of 20% of net profit is paid at 24:00 on 31/12/2017; (4) the same if the dividend of 20% is decided before the end of 2017 and paid on 30/06/2018.

## Phase 1 – Setup: data, timeline and hypotheses

**Period:** calendar year 2017. **Closing:** 31/12/2017. The 2016 data serve only to build the **opening balance** of 2017.

**Hypotheses:**

1. The licences were bought at the very end of 2016, so they start to be amortised in 2017 (no 2016 amortisation is given).
2. The equipment was depreciated only 26 m.u. in 2016 (part of the year); in 2017 it is used the whole year.
3. Interest is paid at the end of each year, so no interest payable remains; the loan is due in 2019, more than one year after 31/12/2017: **long-term** [Slide 16].
4. The financial shares are a long-term investment (non-current asset) [Slide 15]; they produce no revenue in the data.

## Phase 2 – Frames: opening balance and classification of every operation

### Opening balance at 31/12/2016 (rebuilt from the data)

| **Assets** | **m.u.** | **Liabilities and equity** | **m.u.** |
|---|---|---|---|
| Technical equipment (5,371 − 26) | 5,345 | Capital | 6,003 |
| Licences | 206 | Loss of 2016 | (340) |
| Financial shares | 636 | **Equity** | **5,663** |
| Cash | 1,476 | Bank loan (due 2019) | 2,000 |
| **Total** | **7,663** | **Total** | **7,663** |

It balances ($7{,}663 = 5{,}663 + 2{,}000$), which confirms that no other 2016 item is missing.

### Operations of 2017

| **Operation** | **Income statement 2017** | **Cash 2017** | **Open at 31/12/2017** |
|---|---|---|---|
| Subscriptions + on-line sales | Revenue 1,966 | +1,802 | Receivables 164 |
| Purchases of materials (680) | COGS 340 | −560 | Inventory 340; suppliers 120 |
| Staff | 559 | −559 | – |
| Supplies and outside services | 312 | −312 | – |
| Administrative expenses | 300 | −120 | Other creditors 180 |
| Equipment depreciation | 268.55 | 0 | Accumulated depreciation 294.55 |
| Licence amortisation | 41.20 | 0 | Licences 164.80 |
| Interest | 100 | −100 | – |

| Operation | Income statement 2017 | Cash 2017 | Open at 31/12/2017 |
|---|---|---|---|
| Tax | 13.575 | 0 | Taxes payable 13.575 |

## Phase 3 – Derivation

**Step 1. Revenues and receivables**

$$\text{Subscriptions} = 5{,}000 \text{ clients} \times 0.032 \tfrac{\text{m.u.}}{\text{client}\cdot\text{month}} \times 12 \text{ months} = 1{,}920 \text{ m.u.}$$

$$\text{Revenues} = 1{,}920 + 46 = 1{,}966 \text{ m.u.}, \qquad \text{Collected} = 1{,}966 - 164 = 1{,}802 \text{ m.u.}$$

**Step 2. Purchases, consumption and inventory**

Opening stock 0 ("stocks were exhausted at the end of 2016"); purchases 680; half consumed:

$$\text{COGS} = 0 + 680 - 340 = 340 \text{ m.u.}, \qquad \text{Ending inventory} = 340 \text{ m.u.}$$

Payment: 560 paid; $680 - 560 = 120$ owed to suppliers.

**Step 3. Operating expenses**

- Staff 559 and supplies and outside services 312, all paid.
- Administrative expenses 300: 60% deferred, $0.60 \times 300 = 180$ owed; $300 - 180 = 120$ paid.

**Step 4. Depreciation and amortisation**

$$D_{\text{equipment}} = \frac{5{,}371}{20} = 268.55 \text{ m.u.}, \qquad A_{\text{licences}} = \frac{206}{5} = 41.20 \text{ m.u.}$$

Accumulated depreciation of the equipment: $26 + 268.55 = 294.55$; net value $5{,}371 - 294.55 = 5{,}076.45$. Licences: $206 - 41.20 = 164.80$.

**Step 5. Interest**

$$I = 2{,}000 \times 0.05 = 100 \text{ m.u.} \quad \text{(paid at the end of 2017)}$$

**Step 6. Income statement 2017 (requirement 1)**

| Line | Computation | m.u. |
|---|---|---|
| Revenues | $1{,}920 + 46$ | 1,966.000 |
| – Cost of goods sold (materials consumed) | | 340.000 |
| **= Gross profit** | | **1,626.000** |
| – Staff | | 559.000 |
| – Supplies and outside services | | 312.000 |
| – Administrative expenses | | 300.000 |
| – Depreciation and amortisation | $268.55 + 41.20$ | 309.750 |
| **= Income from operations** | $1{,}626 - 1{,}480.75$ | **145.250** |
| – Interest expense | | 100.000 |
| **= Income before taxes** | $145.25 - 100$ | **45.250** |
| – Income taxes (30%) | $0.30 \times 45.25$ | 13.575 |
| **= Net income** | $45.25 - 13.575$ | **31.675** |

**Step 7. Cash at 31/12/2017**

$$\text{Cash} = 1{,}476 + 1{,}802 - 560 - 559 - 312 - 120 - 100 = 1{,}627 \text{ m.u.}$$

**Step 8. Balance sheet at 31/12/2017 (requirement 2)**

All the profit is retained, so it stays in equity and reduces the accumulated loss: $-340 + 31.675 = -308.325$.

| Assets | m.u. | Liabilities and equity | m.u. |
|---|---|---|---|
| Technical equipment | 5,371.000 | Capital | 6,003.000 |
| – Accumulated depreciation | (294.550) | Loss of 2016 | (340.000) |
| Licences (net) | 164.800 | Net income 2017 | 31.675 |
| Financial shares | 636.000 | **Equity** | **5,694.675** |
| **Non-current assets** | **5,877.250** | Bank loan (due 2019) | 2,000.000 |
| Inventory | 340.000 | **Long-term liabilities** | **2,000.000** |

| Assets | m.u. | Liabilities and equity | m.u. |
|---|---|---|---|
| Accounts receivable | 164.000 | Suppliers | 120.000 |
| Cash | 1,627.000 | Other creditors (administration) | 180.000 |
| **Current assets** | **2,131.000** | Taxes payable (Q1 2018) | 13.575 |
| | | **Current liabilities** | **313.575** |
| **Total assets** | **8,008.250** | **Total liabilities and equity** | **8,008.250** |

**Step 9. Dividend paid at 24:00 on 31/12/2017 (requirement 3)**

$$\text{Dividend} = 0.20 \times 31.675 = 6.335 \text{ m.u.}$$

Paying it at 24:00 means cash leaves the company and the profit kept in equity falls by the same amount:

| Item | Before | Change | After |
|---|---|---|---|
| Cash | 1,627.000 | −6.335 | 1,620.665 |
| Equity (net income retained) | 5,694.675 | −6.335 | 5,688.340 |
| Total assets = total liabilities and equity | 8,008.250 | −6.335 | 8,001.915 |

**Step 10. Dividend decided in 2017, paid on 30/06/2018 (requirement 4)**

The dividend is already owed to the shareholders at 31/12/2017 but no cash has left: equity falls and a **current liability** (dividends payable, due within the year) appears.

| Item | Before | Change | After |
|---|---|---|---|
| Equity | 5,694.675 | −6.335 | 5,688.340 |
| Dividends payable (current liability) | 0 | +6.335 | 6.335 |
| Current liabilities | 313.575 | +6.335 | 319.910 |
| Cash and total assets | unchanged | 0 | 8,008.250 |

## Phase 4 – Interpretation and checks

- **Balance:** $5{,}877.25 + 2{,}131 = 8{,}008.25 = 5{,}694.675 + 2{,}000 + 313.575$; requirement 3: $8{,}001.915 = 5{,}688.340 + 2{,}313.575$; requirement 4: $8{,}008.25 = 5{,}688.34 + 2{,}000 + 319.91$.
- **Requirement 3 vs 4:** both reduce equity by the same 6.335; the difference is only on the other side: in 3 an **asset** (cash) falls, in 4 a **liability** rises. A dividend is never an expense, so net income is 31.675 in all three cases.

- **Why the company chose not to pay:** after 2017 the equity is still below the capital (accumulated loss $-308.325$); retaining the profit is self-financing [Slide 23] that rebuilds the equity.
- **Exam traps:** (1) the 2016 data are only the opening balance; (2) inventory is half of the purchases; (3) the licences bought at the end of 2016 are amortised from 2017; (4) the loan due in 2019 is long-term.

### Check against the official key

OFFICIAL SOLUTION, T3_ADDITIONAL_SOLUTIONS.PDF, PP. 16-18

Requirements 1 and 2 coincide with the key: revenues 1,966; cost of goods sold 340; operating expenses 1,511; EBITDA 455; depreciation and amortisation 309.75; EBIT 145.25; interest 100; income before taxes 45.25; tax 13.575; net income 31.675; cash 1,627; total assets 8,008.25; equity 5,694.675. The key rounds to one decimal (309.8; 145.3; 45.3; 13.6; 31.7), which only shifts the last digit.

The key does not answer requirements 3 and 4 (20 % dividend): it only shows "Dividends (@0%)", so Steps 9 and 10 above are not covered by it.

ERRATUM IN THE OFFICIAL SOLUTION

The key gives total liabilities of 2,325.935, but its own items add to $120 + 180 + 13.575 + 2{,}000 = 2{,}313.575$ (2,313.6 rounded), which together with equity of 5,694.7 gives the total of 8,008.3 that it does show. The accumulated depreciation mentions "2009"; the 26 m.u. belong to 2016.

Additional Exercise 4 · Exam type

## T3 Additional Exercise 4 — CEMED, statements 2016 and supplier discount

Original statement · T3_Additional_EXERCISES.pdf, p. 5

CERÁMICAS DEL MEDITERRÁNEO (CEMED) manufactures and distributes ceramic tiles for construction in Valencia.

- CEMED was created on 1/01/2016 by three partners, each contributing 7 m.u., fully in cash.
- On 1/02/2016 it bought a piece of land worth 10 m.u., paying 50% in cash and leaving 50% to be paid one year later. On the same date it bought the factory building, paying 6 m.u. in cash and financing this amount with a mortgage at 5% annual interest for ten years. The factory is amortised over 20 years, linearly, with zero residual value. Each 31 December the company returns 10% of the mortgage principal plus the interest accrued in the year.
- On 1/04/2016 it obtained an additional bank loan of 5 m.u. at 6% annual interest, returning 20% of the principal each semester for the next 2½ years (first repayment on 1/10/2016, second on 1/04/2017, and so on). Interest is paid semiannually, calculated on the amount actually due in that semester, the day after the end of the semester.
- With this loan, on the same date, it bought machinery for 3 m.u. with 60 days payment delay; useful life 10 years, straight-line, zero residual value.
- On 1/07/2016 it acquired 150 tons of raw material at 0.03 m.u./ton. Half of it was used in producing the products sold. 60% of the total cost is paid to the supplier at 90 days; the other 40% remains unpaid at the end of the year.
- Sales revenue: 50 m.u., of which only 70% had been collected at 31/12/2016.
- Personnel plus Social Security: 0.8 m.u., paid in full in 2016.
- Corporation tax: 30%, settled by calendar year and paid on 15/01/2017.
- The company keeps 40% of annual net profits as reserves; the dividends are paid on 31/12/2016.

**Required:** (1) the income statement and the balance sheet at 24:00 on 31/12/2016, explaining the intermediate calculations; (2) the changes in the income statement and the balance sheet if the company had negotiated a 5% discount on the total price of the raw materials in exchange for paying in cash the 60% agreed at 90 days. In which case is the profit higher, and why?

## Phase 1 — Setup: data, timeline and hypotheses

| Date | Event | Cash |
|---|---|---|
| 1/01 | Capital $3 \times 7 = 21$ | +21 |
| 1/02 | Land 10: 50% paid, 50% due 1/02/2017 | −5 |
| 1/02 | Building 6, financed by a mortgage of 6 | +6 −6 |
| 1/04 | Bank loan 5 at 6% | +5 |
| 1/04 | Machinery 3 bought, paid 60 days later (31/05) | −3 (on 31/05) |
| 1/07 | Raw material $150 \times 0.03 = 4.5$ | — |
| 29/09 | 60% of the raw material paid (90 days) | −2.7 |
| 1/10 | Loan: first instalment (20% of 5) and interest of the first semester | −1 −0.15 |
| 31/12 | Mortgage: 10% of principal and interest of the year | −0.6 −0.275 |
| 31/12 | Dividends (60% of net income) | −19.3053 |
| 15/01/2017 | Tax paid (after the closing) | — |

**Hypotheses:**

1. The building is used from 1/02 (11 months) and the machinery from 1/04 (9 months). Land is not depreciated.
2. The mortgage interest of 2016 runs from 1/02 to 31/12 (11 months) on the 6 m.u. outstanding.
3. Loan semesters: 1/04-30/09 (principal 5) and 1/10-31/03 (principal 4). The interest of the second semester is paid on 1/04/2017, so the part from 1/10 to 31/12 is an **accrued, unpaid** expense of 2016.
4. Results to four decimals; exact fractions were used and the last digit is rounded.

## Phase 2 — Frames: classification of every operation

| Operation | Income statement 2016 | Cash 2016 | Open at 31/12/2016 |
|---|---|---|---|
| Capital | — | +21 | Capital 21 |
| Land | — | −5 | Land 10; land payable 5 (current) |
| Building / mortgage | Depreciation 0.275; interest 0.275 | +6 −6 −0.6 −0.275 | Building $6 - 0.275$; mortgage 5.4 (0.6 current, 4.8 long-term) |
| Bank loan | Interest $0.15 + 0.06$ | +5 −1 −0.15 | Loan 4 (2 current, 2 long-term); interest payable 0.06 |
| Machinery | Depreciation 0.225 | −3 | Machinery $3 - 0.225$ |
| Raw material | COGS 2.25 | −2.7 | Inventory 2.25; suppliers 1.8 |
| Sales | Revenue 50 | +35 | Receivables 15 |

| Operation | Income statement 2016 | Cash 2016 | Open at 31/12/2016 |
|---|---|---|---|
| Personnel + SS | 0.8 | -0.8 | – |
| Tax | 13.7895 | 0 | Taxes payable 13.7895 |
| Dividends | – (distribution) | -19.3053 | Reserves 12.8702 |

## Phase 3 – Derivation (requirement 1)

### Step 1. Depreciation

$$D_{\text{building}} = \frac{6-0}{20} \times \frac{11}{12} = 0.3 \times 0.9167 = 0.2750 \text{ m.u.}$$

$$D_{\text{machinery}} = \frac{3-0}{10} \times \frac{9}{12} = 0.3 \times 0.75 = 0.2250 \text{ m.u.}$$

Total depreciation: $0.275 + 0.225 = 0.5$ m.u.

### Step 2. Mortgage

Interest of 2016 (1/02 to 31/12, principal 6):

$$I_{\text{mortgage}} = 6 \times 0.05 \times \frac{11}{12} = 0.2750 \text{ m.u.} \quad (\text{paid on } 31/12)$$

Principal returned on 31/12: $0.10 \times 6 = 0.6$; outstanding $6 - 0.6 = 5.4$. The next 0.6, due 31/12/2017, is current; the remaining $5.4 - 0.6 = 4.8$ is long-term.

### Step 3. Bank loan

Each instalment is $0.20 \times 5 = 1$ m.u. (five instalments: 1/10/2016, 1/04/2017, 1/10/2017, 1/04/2018, 1/10/2018).

- First semester (1/04-30/09, principal 5): $I_1 = 5 \times 0.06 \times \frac{6}{12} = 0.15$, paid on 1/10/2016 together with the first instalment.
- Second semester (1/10/2016-31/03/2017, principal 4): the 2016 part is October-December, $I_2 = 4 \times 0.06 \times \frac{3}{12} = 0.06$, **accrued but unpaid** (it will be paid on 1/04/2017).

Interest of the loan in 2016: $0.15 + 0.06 = 0.21$. Principal outstanding at 31/12: $5 - 1 = 4$, of which the instalments of 1/04/2017 and 1/10/2017 (2) are current and those of 2018 (2) are long-term.

**Step 4. Raw material, cost of goods sold and suppliers**

$$\text{Purchase} = 150 \times 0.03 = 4.5 \text{ m.u.}, \qquad \text{COGS} = \frac{4.5}{2} = 2.25, \qquad \text{Inventory} = 2.25$$

Paid at 90 days: $0.60 \times 4.5 = 2.7$; unpaid: $0.40 \times 4.5 = 1.8$ (suppliers, current).

**Step 5. Sales**

Collected $0.70 \times 50 = 35$; receivables $50 - 35 = 15$.

**Step 6. Income statement 2016**

| Line | Computation | m.u. |
|---|---|---|
| Sales revenue | | 50.0000 |
| – Cost of goods sold | Step 4 | 2.2500 |
| **= Gross profit** | | **47.7500** |
| – Personnel and Social Security | | 0.8000 |
| – Depreciation | Step 1 | 0.5000 |
| **= Income from operations** | $47.75 - 1.3$ | **46.4500** |
| – Interest (mortgage 0.275; loan 0.21) | Steps 2-3 | 0.4850 |
| **= Income before taxes** | $46.45 - 0.485$ | **45.9650** |
| – Income taxes (30%) | $0.30 \times 45.965$ | 13.7895 |
| **= Net income** | $45.965 - 13.7895$ | **32.1755** |

Distribution: reserves $= 0.40 \times 32.1755 = 12.8702$; dividends $= 0.60 \times 32.1755 = 19.3053$, paid on 31/12.

**Step 7. Cash at 24:00 on 31/12/2016**

| Inflows | m.u. | Outflows | m.u. |
|---|---|---|---|
| Capital | 21.0000 | Land (50%) | 5.0000 |
| Mortgage | 6.0000 | Building | 6.0000 |
| Bank loan | 5.0000 | Machinery | 3.0000 |
| Sales collected | 35.0000 | Raw material (60%) | 2.7000 |
| | | Personnel | 0.8000 |
| | | Mortgage principal + interest | 0.8750 |
| | | Loan instalment + interest | 1.1500 |
| | | Dividends | 19.3053 |
| **Total** | **67.0000** | **Total** | **38.8303** |

$$\text{Cash} = 67 - 38.8303 = 28.1697 \text{ m.u.}$$

**Step 8. Balance sheet at 24:00 on 31/12/2016**

| **Assets** | **m.u.** | **Liabilities and equity** | **m.u.** |
|---|---|---|---|
| Land | 10.0000 | Capital | 21.0000 |
| Building ($6 - 0.275$) | 5.7250 | Reserves (40% of net income) | 12.8702 |
| Machinery ($3 - 0.225$) | 2.7750 | **Equity** | **33.8702** |
| **Non-current assets** | **18.5000** | Mortgage, long-term part | 4.8000 |
| Inventory (raw material) | 2.2500 | Bank loan, long-term part | 2.0000 |
| Accounts receivable | 15.0000 | **Long-term liabilities** | **6.8000** |
| Cash | 28.1697 | Land payable (1/02/2017) | 5.0000 |
| **Current assets** | **45.4197** | Mortgage, current part | 0.6000 |
| | | Bank loan, current part | 2.0000 |
| | | Interest payable | 0.0600 |
| | | Suppliers | 1.8000 |
| | | Taxes payable (15/01/2017) | 13.7895 |
| | | **Current liabilities** | **23.2495** |
| **Total assets** | **63.9197** | **Total liabilities and equity** | **63.9197** |

## Phase 3b – Derivation (requirement 2: 5% discount, 60% paid in cash)

**Step 9. New purchase price and payments**

$$\text{Price} = 4.5 \times (1 - 0.05) = 4.275 \text{ m.u.}$$

- Paid in cash on 1/07: $0.60 \times 4.275 = 2.565$ (instead of 2.7 at 90 days).
- Unpaid at 31/12: $0.40 \times 4.275 = 1.71$ (instead of 1.8).
- Consumed (half): $\text{COGS} = 4.275/2 = 2.1375$; inventory 2.1375 (both instead of 2.25).

**Step 10. New income statement**

Only the cost of goods sold changes before taxes: $\Delta\text{COGS} = 2.1375 - 2.25 = -0.1125$.

| Line | Requirement 1 | Requirement 2 | Change |
|---|---:|---:|---:|
| Cost of goods sold | 2.2500 | 2.1375 | −0.1125 |
| Income before taxes | 45.9650 | 46.0775 | +0.1125 |
| Income taxes (30%) | 13.7895 | 13.8233 | +0.0338 |
| **Net income** | **32.1755** | **32.2543** | **+0.0788** |
| Dividends (60%) | 19.3053 | 19.3526 | +0.0473 |
| Reserves (40%) | 12.8702 | 12.9017 | +0.0315 |

The net change is the after-tax saving: $0.1125 \times (1 - 0.30) = 0.07875$.

**Step 11. New balance sheet items**

| Item | Change | Reason |
|---|---:|---|
| Inventory | −0.1125 | Half of the material is still in stock, valued at the lower price |
| Cash | +0.0878 | Pays 2.565 instead of 2.7 (+0.135) but pays 0.0473 more dividends |
| Suppliers | −0.0900 | 40% of a lower price (1.71 instead of 1.8) |
| Taxes payable | +0.0338 | Higher profit |
| Reserves | +0.0315 | 40% of the higher net income |
| **Total assets** | **−0.0248** | $-0.1125 + 0.08775$ |
| **Total liabilities and equity** | **−0.0248** | $-0.09 + 0.03375 + 0.0315$ |

New totals: $63.9197 - 0.0248 = 63.8950$ on both sides (cash 28.2575).

## Phase 4 – Interpretation and checks

- **Balance:** $18.5 + 45.4197 = 63.9197 = 33.8702 + 6.8 + 23.2495$; requirement 2: both sides fall by 0.02475.
- **In which case is the profit higher?** With the **discount**. The discount lowers the cost of the material; the price paid for it is giving up 90 days of **trade credit** [Slide 35], and trade credit has no explicit interest in this problem, so losing it costs nothing in the income statement: it only advances a payment that in both cases falls inside 2016.
- **Why only half of the saving reaches the profit:** only half of the material was consumed. The other half of the saving is in the lower value of the inventory and will reach the income statement when that stock is used.
- **Debt structure** [Slide 36]: liabilities $6.8 + 23.2495 = 30.0495$, equity 33.8702, so $L/E = 0.89$.
- **Exam traps:** (1) the mortgage finances the building: cash +6 and −6 on the same day; (2) the loan interest of October-December is accrued but unpaid; (3) the current part of each loan is what is due before 31/12/2017; (4) the land is not depreciated; (5) taxes are payable on 15/01/2017, so they are a liability, while dividends were paid.

## Check against the official key

OFFICIAL SOLUTION, T3_ADDITIONAL_SOLUTIONS.PDF, PP. 19-25

Requirement 1 coincides with the key: cost of goods sold 2.25; personnel 0.8; depreciation 0.5; EBIT 46.45; interest 0.485; income before taxes 45.965; tax 13.79; net income 32.18; dividends 19.31; reserves 12.87; cash 28.17; total 63.92. Requirement 2 also coincides: cost of goods sold 2.14 (2.1375, rounded in the key); net income 32.25; dividends 19.35; cash 28.26; suppliers 1.71; total 63.89. The key's figures show net income rising from 32.18 to 32.25, but the key states no conclusion to the question "in which case are profits higher?"; the reasoning in Phase 4 is ours.

DISCREPANCY WITH THE OFFICIAL SOLUTION (CURRENT PART OF THE DEBT)

The key leaves the whole bank loan (4.0) and the whole mortgage (5.4) as long-term debt, so its current liabilities are 20.65 and its long-term liabilities 9.4. Here the instalments due within twelve months of 31/12/2016 are shown as current: 2.0 of the loan (1/04/2017 and 1/10/2017) and 0.6 of the mortgage (31/12/2017), following the definition of Slide 16. Current liabilities are therefore 23.25 and long-term liabilities 6.8; total liabilities (30.05), equity and total assets are the same.

DIFFERENCES IN THE WORDING OF THE KEY

The key's statement dates the company in 2010, while the official sheet in `sources/` says 2016, which is the year used here. In the key's part 2 the interest formula writes 0.85 where 0.80 is meant (the loan balance of 4 over 5); its value, 0.06, is unchanged.